

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 7: Cloud Classification (10 Genera) & Fog Types

Syllabus Source: CAE Oxford (pp. 209-248) | Pass Target: >= 90% AviationExam

1. THE 10 CLOUD GENERA (ICAO CLASSIFICATION)

Clouds are classified by height of base and physical appearance into 10 official genera:

Cloud Level & Height Range Official Cloud Genera Codes		Physical Characteristics & Composition
Cloud Level & Height Range	Official Cloud Genera Codes	Physical Characteristics & Composition
High Clouds (Base > 20,000 ft / FL 200)	Ci (Cirrus), Cc (Cirrocumulus), Cs (Cirrostratus)	Composed purely of ICE CRYSTALS. Cs causes optical HALO around sun/moon. Ci mares' tails signal approaching warm front.
Medium Clouds (Base 6,500 to 20,000 ft)	Ac (Alto cumulus), As (Altostratus), Ns (Nimbostratus)	Mixed water droplets and ice. As gives watery sun appearance. Ns produces continuous rain/snow (dark and thick).
Low Clouds (Base 0 to 6,500 ft AGL)	Sc (Strato cumulus), St (Stratus), Cu (Cumulus)	Water droplets. St is a uniform grey layer causing drizzle. Sc is a rolling grey sheet.
Clouds with Vertical Development	Cb (Cumulonimbus), TCU (Towering Cumulus)	Massive vertical anvil cloud. Generates violent thunderstorms, severe icing, hail, lightning, and microbursts!

2. FOG FORMATION MECHANISMS & DIFFERENCES

- **Radiation Fog:** Formed over land on CLEAR, CALM NIGHTS with light wind (2 to 8 kt) and high RH. Terrestrial radiation chills surface air below dew point. Disperses mid-morning with solar heating or strong wind (> 10 kt mixes and lifts it into Stratus). NEVER OCCURS OVER THE OPEN SEA!
- **Advection Fog:** Formed when WARM, MOIST AIR moves over a COLDER surface. Requires moderate wind (10 to 15 kt). Extremely persistent day and night, over both land and sea (e.g. San Francisco, English Channel).
- **Upslope Fog (Hill Fog):** Formed when moist air is forced up mountain slopes, cooling adiabatically to its dew point.
- **Steam Fog (Arctic Sea Smoke):** Formed when extremely COLD, DRY AIR blows over WARM WATER. Water evaporates rapidly into the cold air, instantly condensing like steam.

3. FOG VS MIST VS HAZE

- **Fog (FG):** Visibility < 1,000 meters, Relative Humidity ~ 100% (water droplets).
- **Mist (BR):** Visibility 1,000 to 5,000 meters, Relative Humidity >= 95% (water droplets).
- **Haze (HZ):** Visibility <= 5,000 meters, Relative Humidity < 95% (dry dust, smoke, pollutants).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Diagnosing Radiation Fog vs Advection Fog

Radiation Fog Diagnosis: - Requires: Clear skies at night, moist soil, light wind of 2 to 8 kt (5 kt ideal). - Calm wind (0 kt) produces dew/frost only; wind > 10 kt mixes air into low stratus. - Dissipates with morning solar radiation. Advection Fog Diagnosis: - Requires: Warm, moist air blowing over a significantly colder land or sea surface. - Requires wind of 10 to 15 kt. Can form day or night, over sea or coast. - Will NOT burn off with morning sunlight; requires a change of air mass.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Which of the following conditions are REQUIRED for the formation of RADIATION FOG?**

- [A]: Strong wind and thick overcast cloud
- [B]: A CLEAR SKY at night, high relative humidity, and LIGHT WIND (2 to 8 knots)
- [C]: Warm moist air blowing over cold ocean currents
- [D]: Freezing rain

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Radiation fog requires maximum terrestrial cooling (clear sky) and light breeze (2-8 kt) to mix the cooled air throughout the shallow surface layer without dispersing it.

Q2: Why can RADIATION FOG never form over the open ocean?

- [A]: Sea water contains salt
- [B]: The sea surface temperature undergoes virtually NO DIURNAL TEMPERATURE CHANGE (only ~0.5°C to 1°C)
- [C]: Winds over the sea are always calm
- [D]: Too much evaporation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The high specific heat capacity of water prevents the sea surface from cooling rapidly at night, making radiation fog impossible over the open sea.

Q3: Which cloud genus is responsible for producing an optical HALO around the sun or moon?

- [A]: Altostratus
- [B]: CIRROSTRATUS (Cs), due to light refraction through hexagonal ice crystals
- [C]: Stratocumulus
- [D]: Nimbostratus

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cirrostratus is composed entirely of ice crystals that act as prisms, refracting sunlight or moonlight at a 22° angle to produce a prominent halo.